INFOTEC
Centro de Investigación e Innovación en TIC

Modelación probabilística de periodos de siembra del maíz en condiciones de cambio climático. Caso de estudio en Alemania.

PRESENTA

Isaac Vázquez Mendoza

BAJO LA DIRECCIÓN DE

Dra. Magali Arellano Vázquez

Motivación del problema

Invierno

Verano

Primavera

Otoño

Motivación del problema

Invierno

Verano

Primavera

Otoño

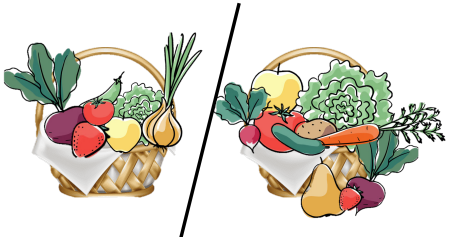
Motivación del problema

Invierno

Verano

Primavera

Otoño



Movilización en fechas de siembra

Adelantar

- Reduce el estrés asociado a temporadas más cálidas.
- Incrementa la cosecha.
- Retrasa la etapa de emergencia.
- Reduce la densidad poblacional del cultivo.

Fuente: Clark, *et al.* (2026); Maresma *et al.* (2019); Mason *et al.* (2018)

Mantener

- Resiliencia de la planta.
- Emplear métodos externos.

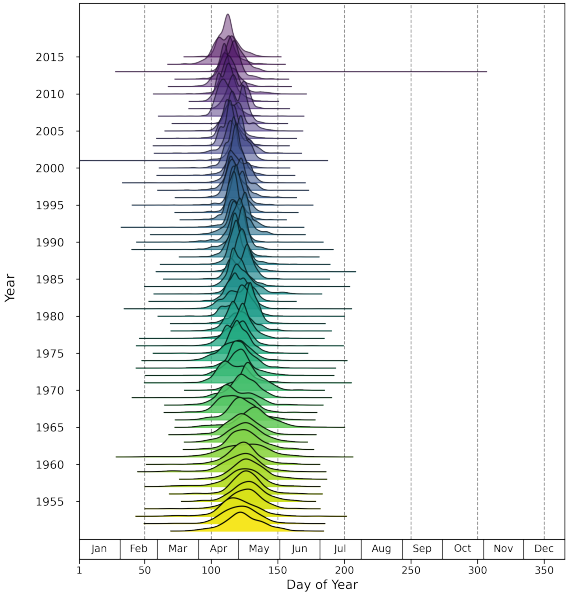
Fuente: Agrama, H. (1996); Mason *et al.* (2018); Wang *et al.* (2025).

Retrasar

- Ofrece condiciones más cálidas.
- Acelera la maduración.
- Aumenta la humedad de los granos.
- Incrementa la altura de las plantas.
- Reduce la producción del cultivo.

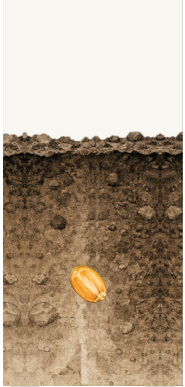
Fuente: Clark *et al.* (2026); Maresma *et al.* (2019); Donohue *et al.* (2015).

Movilización de fechas de siembra en Alemania



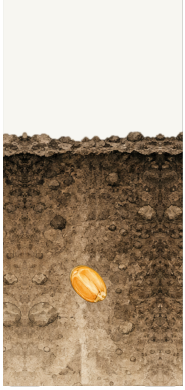
- Fechas de siembra tomadas de la plataforma Proyecto Fenología Paneuropea 725 (PEP725).
- Periodo 1951–2015.
- Tamaño de la muestra: 58115.

¿Cómo determinar los periodos de siembra?



Siembra

Emergencia



Siembra

Emergencia

16.6 ± 3.1 días.

Fuente: Chmielewski *et al.* (2004); Sánchez *et al.* (2014).

Maíz en Alemania

Europa cuenta con el 8.91% de las zonas de producción.

Alemania tendrá un incremento en las zonas con potencial de producción agrícola del maíz.

Fuente: Miedaner *et al.* (2021).

Ante las alteraciones en los procesos agrícolas derivadas del cambio climático, ¿cómo predecir ventanas de siembra de maíz en el sureste de Alemania mediante modelos probabilísticos empleando datos agroclimáticos más allá de los GDD?

Objetivos generales

Desarrollar y validar un modelo probabilístico robusto, independiente de los GDD, que integre variables agroclimáticas significativas para predecir ventanas de fechas de siembra de maíz en el sureste de Alemania.

Objetivos generales

Desarrollar y validar un modelo probabilístico robusto, independiente de los GDD, que integre variables agroclimáticas significativas para predecir ventanas de fechas de siembra de maíz en el sureste de Alemania.

Favorecer los ciclos de desarrollo

Objetivos generales

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Favorecer los ciclos de desarrollo	Disminuir la exposición a estresores ambientales

Objetivos generales

Desarrollar y validar un modelo probabilístico robusto, independiente de los GDD, que integre variables agroclimáticas significativas para predecir ventanas de fechas de siembra de maíz en el sureste de Alemania.

Favorecer los ciclos de desarrollo	Disminuir la exposición a estresores ambientales	Contribuir a la toma de decisiones

Objetivos generales

Desarrollar y validar un modelo probabilístico robusto, independiente de los GDD, que integre variables agroclimáticas significativas para predecir ventanas de fechas de siembra de maíz en el sureste de Alemania.

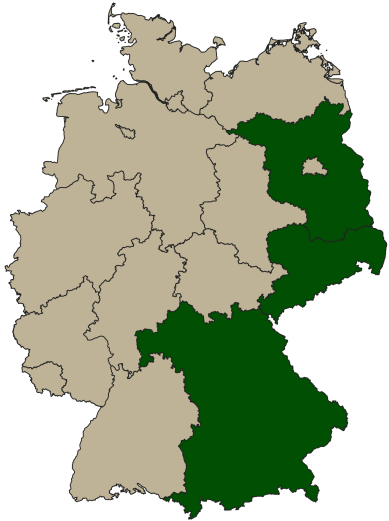
Favorecer los ciclos de desarrollo	Disminuir la exposición a estresores ambientales	Contribuir a la toma de decisiones

Contrastándolo con información oficial reportada para Alemania, un país con potencial agrícola en crecimiento debido a que se enfrenta a un contexto de transición climática derivado del cambio climático.

Datos agroclimáticos

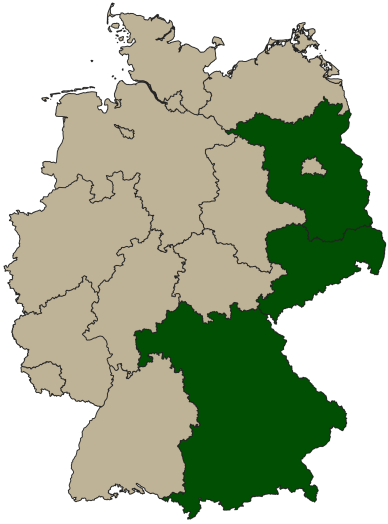
- Temperatura ambiente promedio.
- Humedad relativa.
- Temperatura del suelo (5 cm de profundidad).
- Humedad del suelo disponible para el cultivo del maíz (0–10 cm de profundidad).

Fuente: Deutscher Wetterdienst (2025).



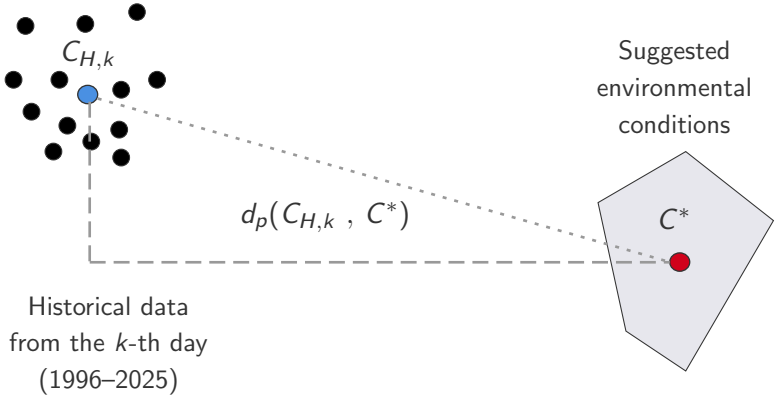
Datos agroclimáticos

- Temperatura ambiente promedio. (11.10 ± 2.5116 °C.)
- Humedad relativa. (65–85 %.)
- Temperatura del suelo (5 cm de profundidad). (8–10 ° C.)
- Humedad del suelo disponible para el cultivo del maíz (0–10 cm de profundidad). (80–90%.)

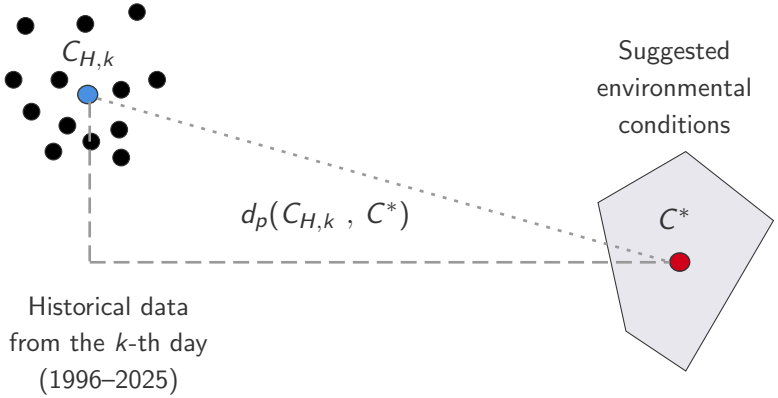


Fuente: Deutscher Wetterdienst (2025).

Probabilidades de transición

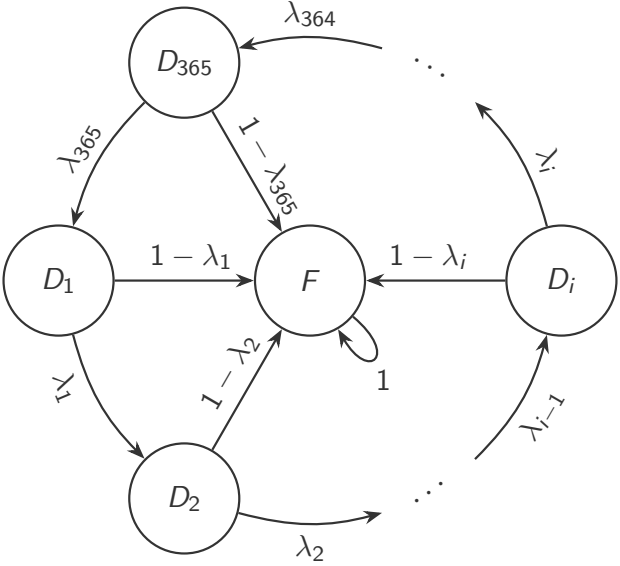


Probabilidades de transición

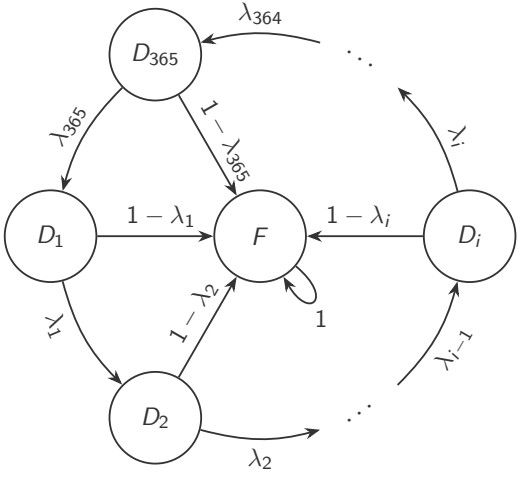


$$\lambda_k := 1 - \frac{d(C_{H,k}\ ,\ C^*)}{\max\limits_{l\in\{1,\ldots,365\}} \{d(C_{H,l}\ ,\ C^*)\}}.$$

Modelo


λ_{364}
D_{365}
$\ddots$
λ_{365}
$1-\lambda_{365}$
λ_i
D_1
$1-\lambda_1$
F
$1-\lambda_i$
D_i
λ_1
$1-\lambda_2$
1
λ_{i-1}
D_2
$\cdots$
λ_2

Modelo: Criterio empírico de permanencia



Calcular

$$P_i(\tau_F > n^*) := P(\tau_F > n^* \mid X_0 = D_i),$$

con $\{\tau_F > n^*\} = \inf_{t \in \mathbb{N}} \{X_t = F\}$.

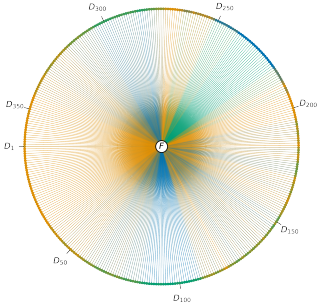
Metodología de de Anda-Jauregui *et al.* (2021).

Por realizar $r^* \in \mathbb{N}$ caminatas aleatorias

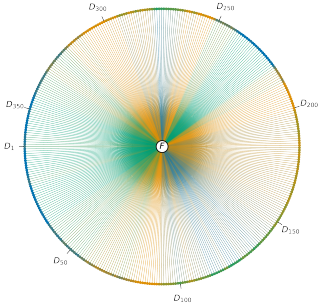
$$\widehat{P}_i(\tau_F > n^*) := \frac{|W(D_i, D_{(i+n^*) \bmod 365})|}{r^*} \geq \zeta^*.$$

Intervalos de $(1-\alpha)100\%$ de confianza por bootstrap y jackknife.

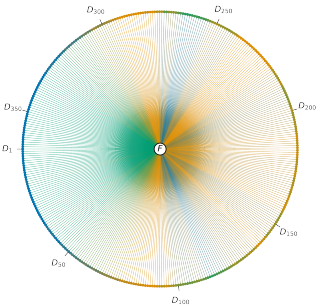
Resultados: Gráficas dirigidas con pesos



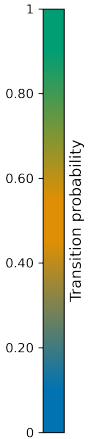
Brandemburgo



Sajonia

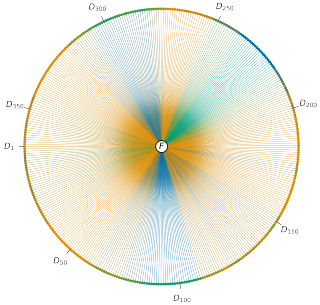


Baviera

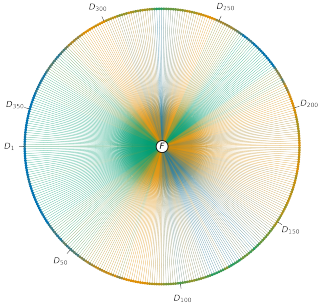


Distancia Euclidiana

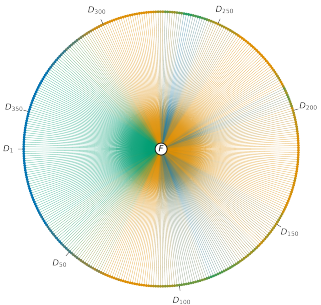
Resultados: Gráficas dirigidas con pesos



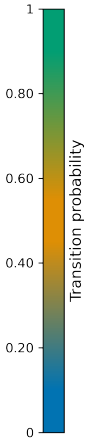
Brandemburgo



Sajonia

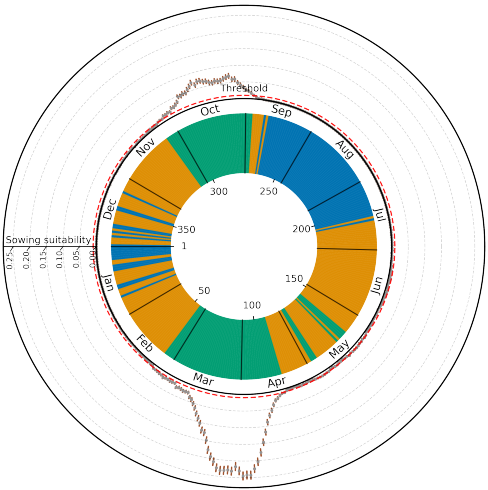


Baviera

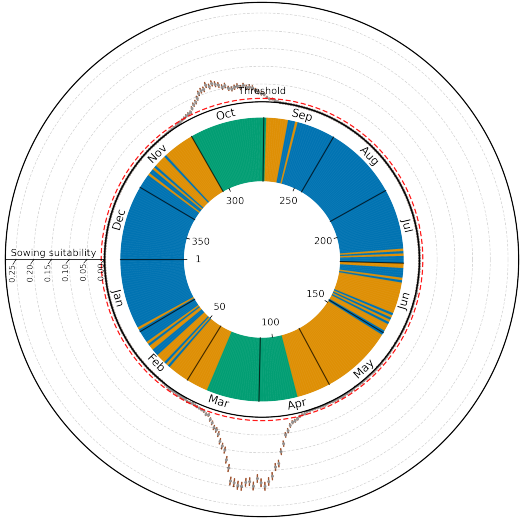


Distancia Manhattan

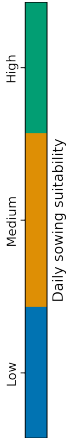
Resultados: Brandemburgo



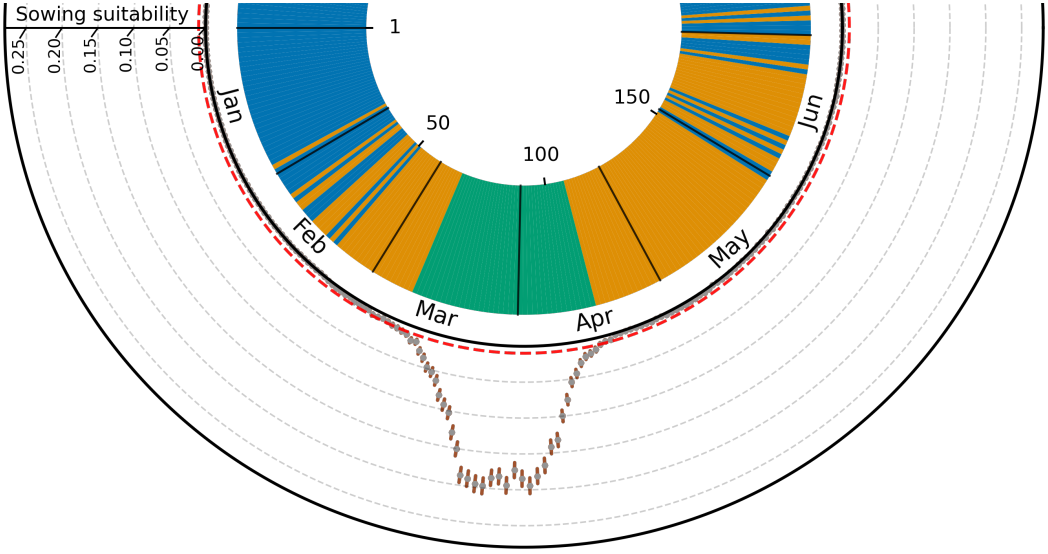
Distancia Euclidiana



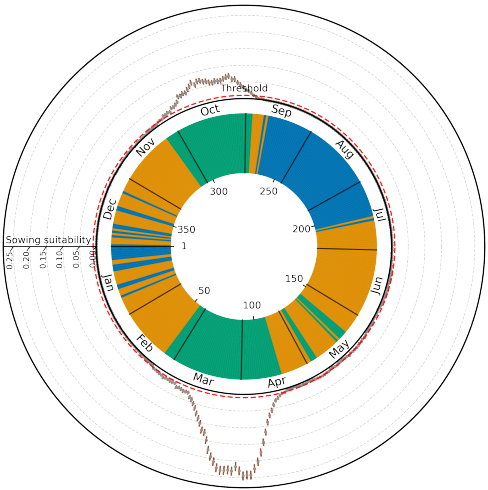
Distancia Manhattan



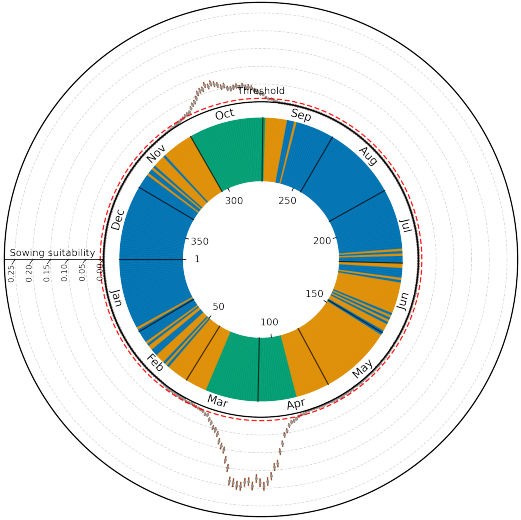
Resultados: Brandemburgo



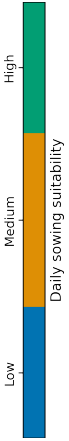
Resultados: Brandemburgo



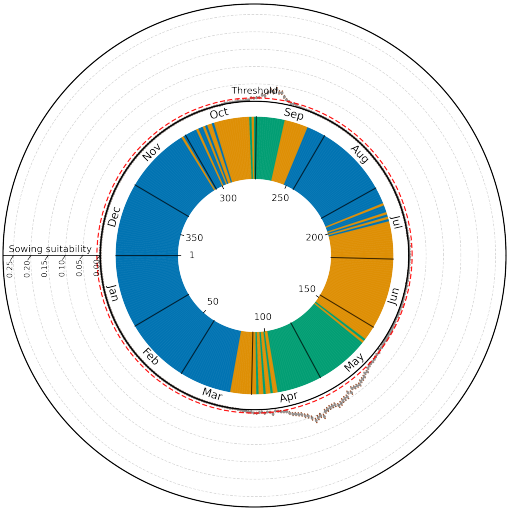
Distancia Euclidiana



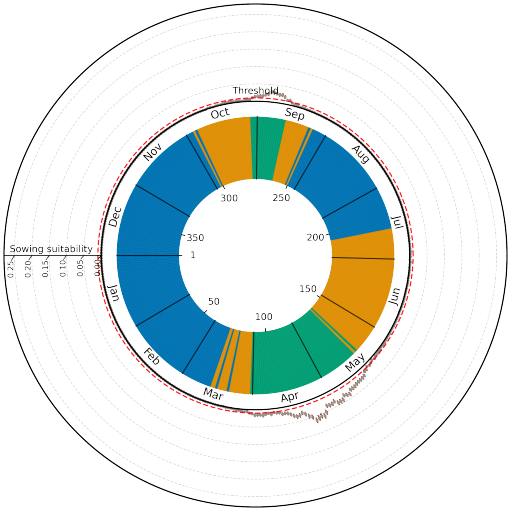
Distancia Manhattan



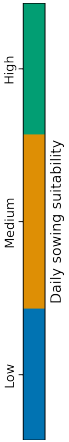
Resultados: Sajonia



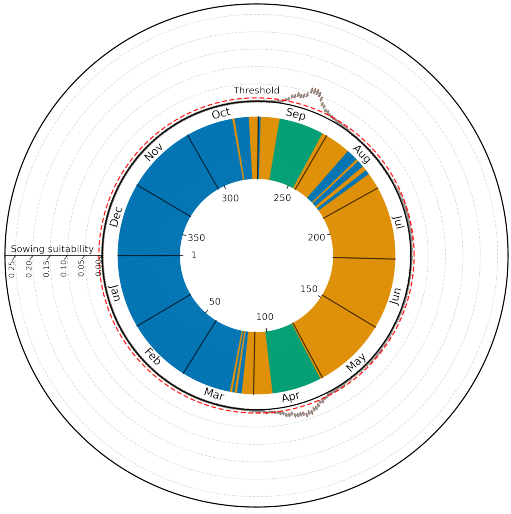
Distancia Euclidiana



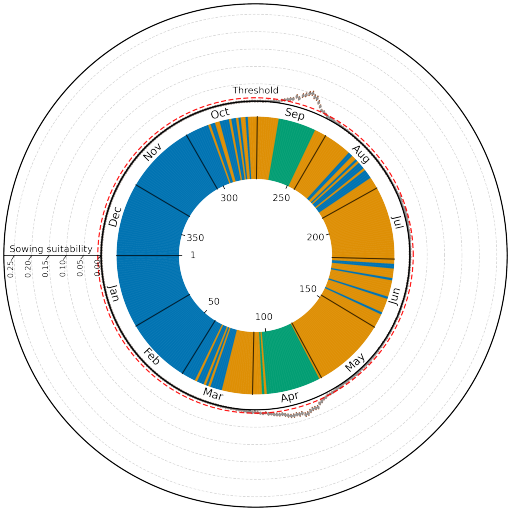
Distancia Manhattan



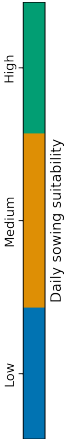
Resultados: Baviera



Distancia Euclidiana



Distancia Manhattan



Benchmarks

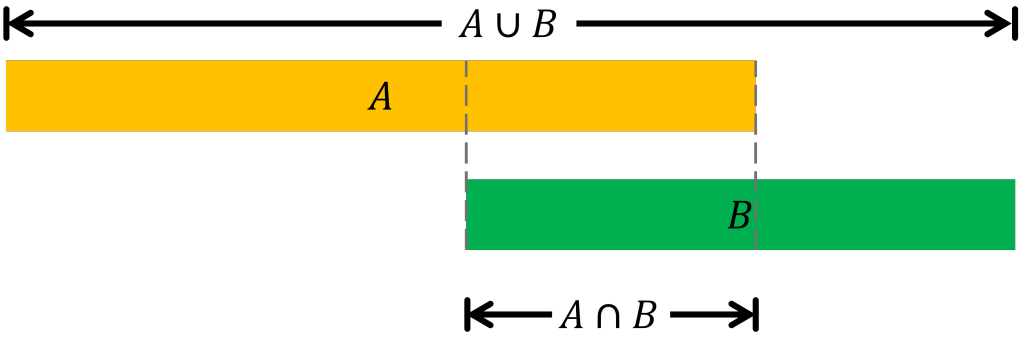
Región	Características	Fecha de siembra
Brandemburgo Norte de Sajonia	Altitud menor que 250 m.s.n.m.	115.6 ± 7.12
	Latitud mayor o igual que 51° N.	116.2 ± 7.08
	Longitud mayor o igual que 8.5° E.	116.2 ± 7.3
Sur de Sajonia Baviera	Altitud mayor o igual que 250 m.s.n.m.	117.4 ± 8.29
	Latitud menor que 51° N.	116.7 ± 8.23
	Longitud mayor o igual que 8.5° E.	116.2 ± 7.3

Fuente: Parker *et al.* (2017); Flores *et al.* (2025).

	Histórico (1975–2010)	**RCP 4.5 (2050)**	**RCP 8.5 (2050)**
Periodo de siembra	79–120	48–110	38–107

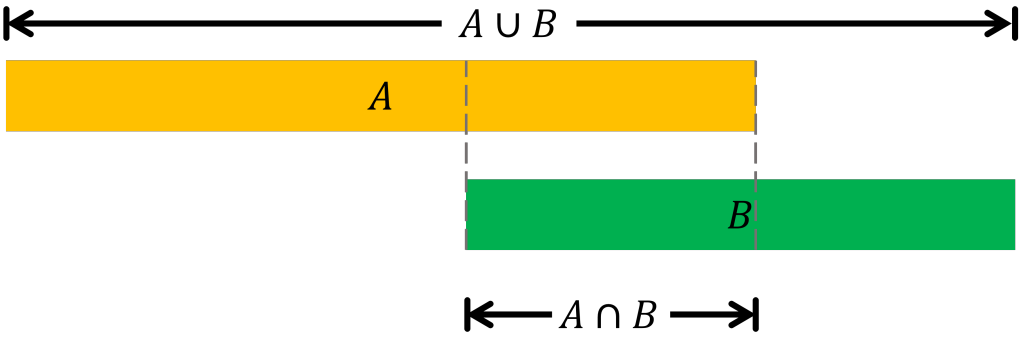
Fuente: Parent *et al.* (2018).

Comparaciones: Índice de Jaccard



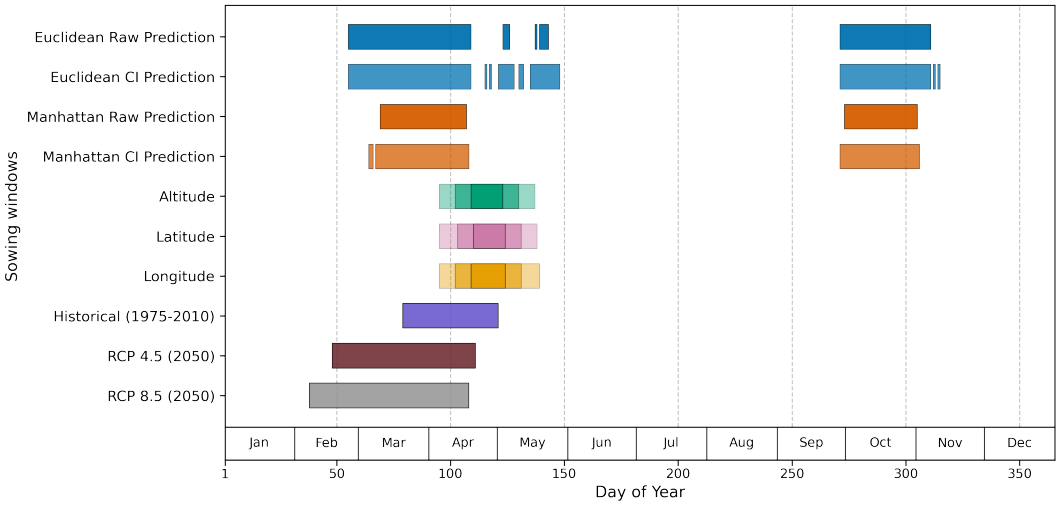
$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}.$$

Comparaciones: Índice de Jaccard

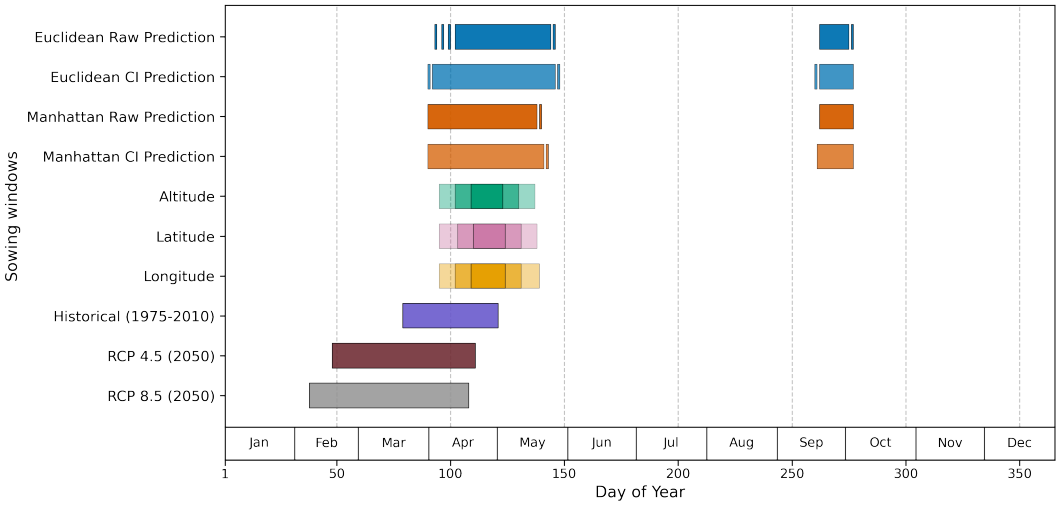


$$0 \leq J(A,B) = \frac{|A \cap B|}{|A \cup B|} \leq 1.$$

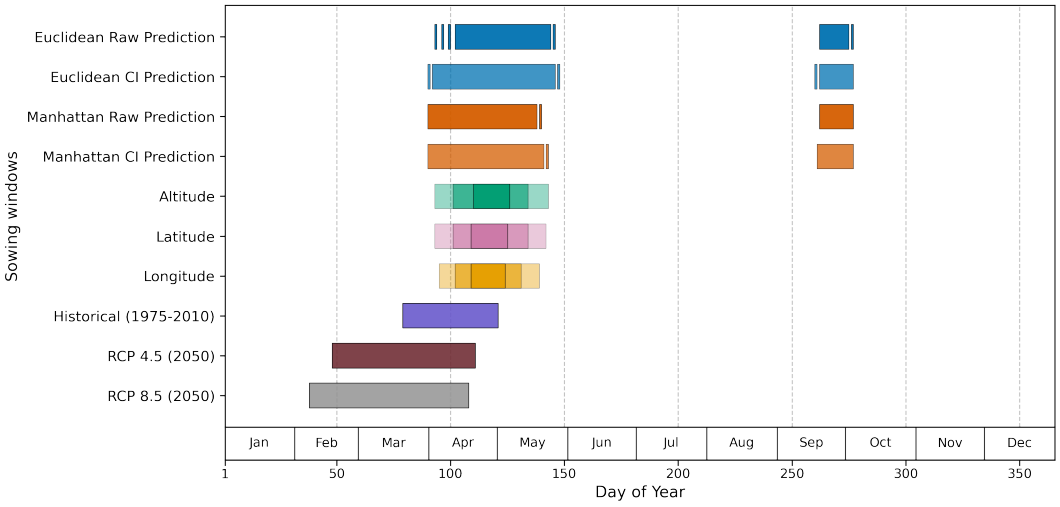
Comparaciones: Brandemburgo



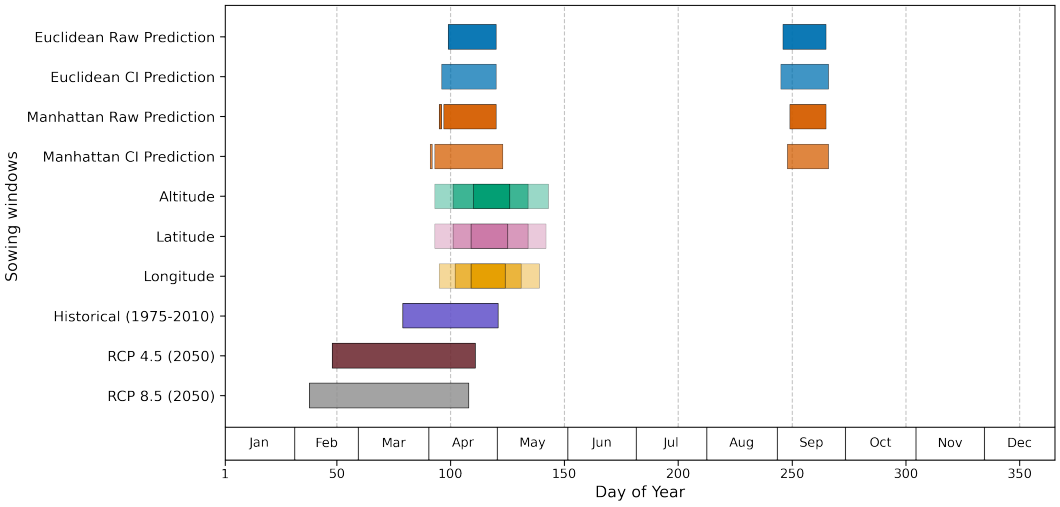
Comparaciones: Norte de Sajonia



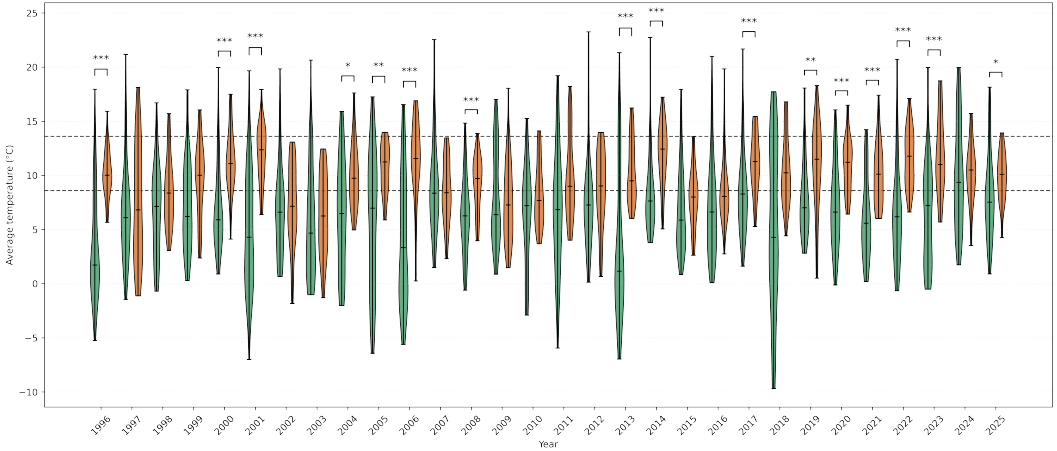
Comparaciones: Sur de Sajonia



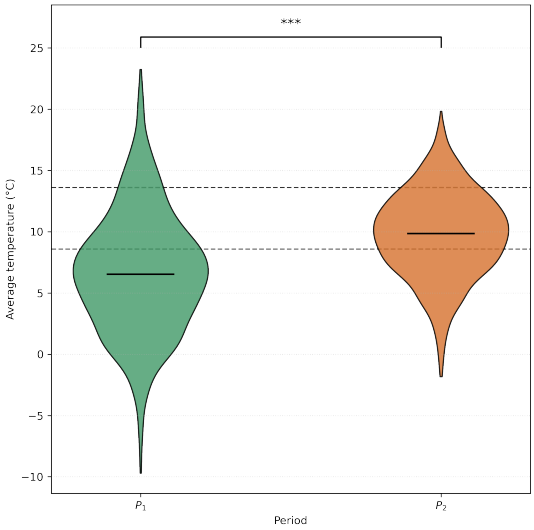
Comparaciones: Baviera



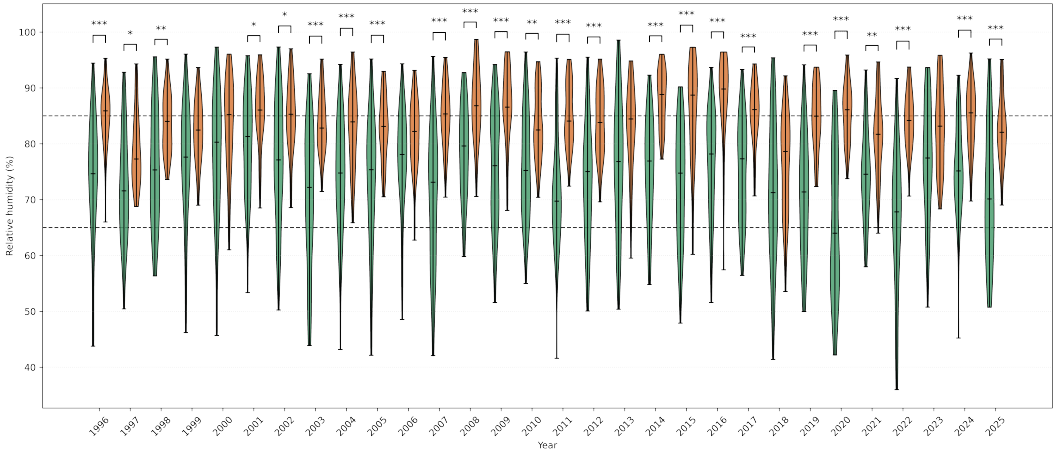
Comparaciones intragrupales desagregadas: Temperatura ambiente



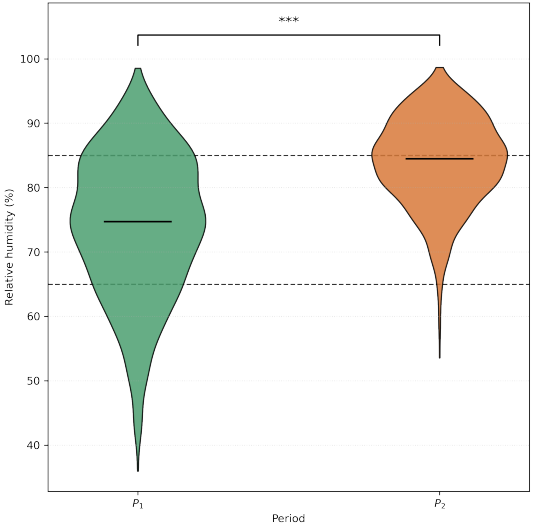
Comparaciones intragrupales agregadas: Temperatura ambiente



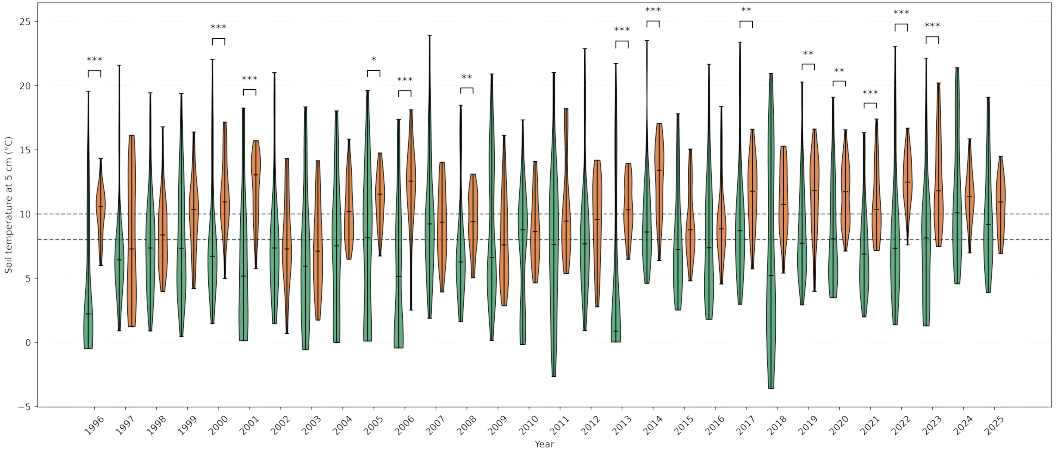
Comparaciones intragrupales desagregadas: Humedad relativa



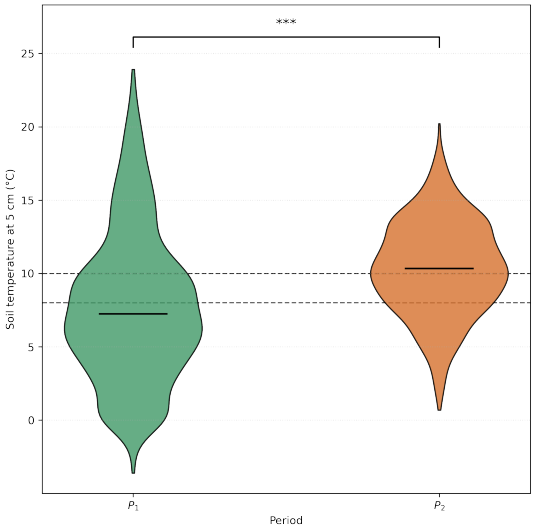
Comparaciones intragrupales agregadas: Humedad relativa



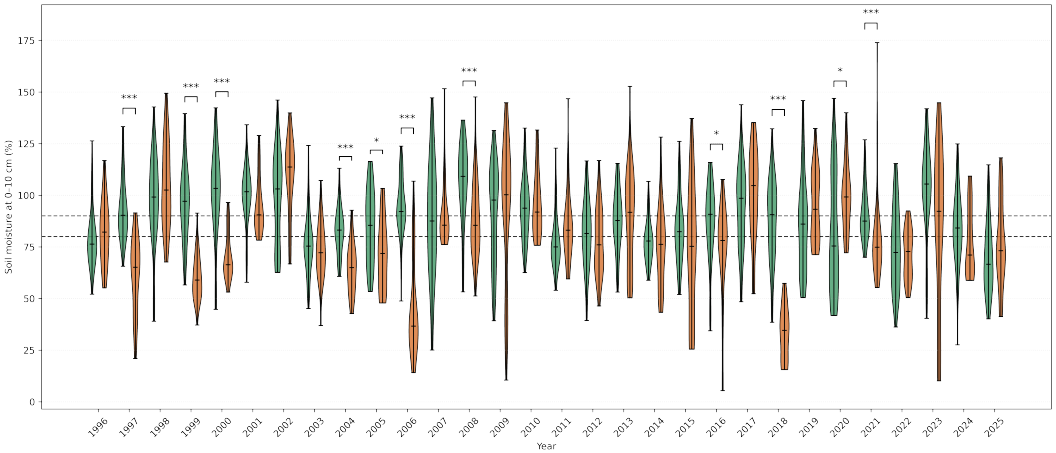
Comparaciones intragrupales desagregadas: Temperatura del suelo



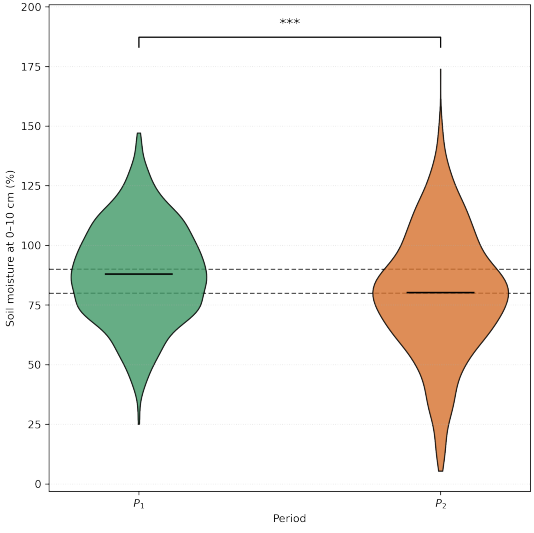
Comparaciones intragrupales agregadas: Temperatura del suelo



Comparaciones intragrupales desagregadas: Humedad del suelo



Comparaciones intragrupales agregadas: Humedad del suelo



INTERNATIONAL STATISTICAL ECOLOGY CONFERENCE 2027

MODELING SOWING TIMING UNDER CLIMATE VARIABILITY USING RANDOM WALKS ON GRAPHS

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Introduction

The progression and duration of crop developmental stages are largely driven by environmental conditions [9]. Climate change is expected to alter the timing and duration of these stages [13, 12]. According to Baum *et al.* (2015) [2], sowing dates may account for up to 70% of maize yield variability. Thus, understanding how climate change may influence the timing of sowing is crucial to shift earlier or later in the year [10]. Climate change projections suggest that northern European regions, including Germany, may become increasingly suitable for maize cultivation [11]. In Germany, maize sowing dates have shifted towards earlier dates, as illustrated in Figure 1b. This shift has been associated with climate change and has intensified since the late 1980s [3]. Traditionally, phenological event timing is predicted using Growing Degree Days (GDD), an approach based primarily on air temperature [1]. In this context, we developed a graph-based random walk model to predict the sowing window for maize in Germany. Two human federal states considered in this study: Bavaria, Brandenburg, and Saxony, see Figure 1a. A region encompassing approximately 35% of the country's area devoted to major crops [4].

(a) Area of study. From top to bottom: Brandenburg, Saxony, and Bavaria.

(b) Distributions of maize sowing dates in Germany.

Figure 1: Geographical context and historical trends in maize sowing dates in Germany.

Methodology

Four raster datasets obtained from the Deutscher Wetterdienst (DWD) [5, 6, 7, 8] covering the period 1996–2025, were used as input data for the probabilistic model. These datasets provide daily measurements of mean air temperature, mean relative humidity, mean soil temperature at 5 cm, and mean soil moisture at 10–50 cm depth, all at a spatial resolution of approximately 1 km^2 (1 km grid). The literature review established the thresholds for each variable were given, see Table 1. For each day t, $x \in \{1, \ldots, 365\}$, a cloud of points in $\mathbb{R}^2$ is constructed by gathering the data corresponding to the t-th day of each year within the period under consideration. Thus, the goodness-of-sowing of day t can be assessed by computing the distance between the centroid of this point cloud, $\bar{c}_{t,(2)}$, and the centroid of the region defined as the Cartesian product of the intervals reported in Table 1, C. This region represents the suggested environmental conditions for maize sowing.

Agroclimatic variable	Interval used in this study
Mean air temperature	11.2 ± 2.2 °C
Mean relative humidity	65-85 %
Mean soil temperature at a depth of 5 cm	8–10 °C
Mean soil moisture in the 0–10 cm soil layer	80–90 %

Table 1: Base threshold per agroclimatic variable used.

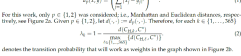
Recalling that given $\mathbf{x} = (x_1, \ldots, x_n), \mathbf{y} = (y_1, \ldots, y_n) \in \mathbb{R}^n$ and $p \in \mathbb{R}$, $p \geq 1$, the p-distance between $\mathbf{x}$ and $\mathbf{y}$ is given by

$$d_p(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p} \tag{1}$$

For this work, only $p \in \{1, 2\}$ was considered, i.e., Manhattan and Euclidean distances, respectively, see Figure 2a. Given $p \in \{1, 2\}$, let $d_i(\cdot, \cdot) = d_p(\cdot, \cdot)$. Therefore, for each $k \in \{1, \ldots, 365\}$

$$\lambda_k = 1 - \frac{d(\bar{c}_{(2),k}, C)}{\max_{1 \leq t \leq 365} \{d(\bar{c}_{(2),t}, C)\}} \tag{2}$$

denotes the transition probability that will work as weights in the graph shown in Figure 2b.



(a) Goodness-of-sowing of day $k, k \in \{1, \ldots, 365\}$ measured by Directed graph with weights modeling the sowing process.

Figure 2: From centroid distances to a weighted directed graph for sowing suitability.

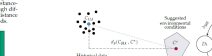
In Figure 2b, vertices $\mathcal{D}_i$ represent the days of the year (DOY), whereas F represents failure. Let $X = \{X_n\}_{n \geq 0}$ be a random walk on this directed graph. If $X_0 = \mathcal{D}_i$, for some $i \in \{1, \ldots, 365\}$, a viable maize seed is considered to have been sown on the i-th DOY. Otherwise, if $X_0 = F$, the seed is considered non-viable. For $t \geq 1$, if $X_t = \mathcal{D}_i$ for some $i \in \{1, \ldots, 365\}$, the germination process is considered successful up to the i-th DOY, whereas $X_t = F$ indicates germination failure due to unfavorable environmental conditions. Furthermore, if $X_t = \mathcal{D}_i$, the process continues to the next DOY provided that

$$u_t \geq \lambda_{[i-1 \bmod 365]} \tag{3}$$

where u_t is a realization of $\mathcal{U} \sim \text{Uniform}(0, 1)$. Otherwise, the process terminates, i.e., $X_j = F$, for every $j \geq t$. Random walks were generated and simulated for each DOY. In Germany, maize germination takes 3 to ± 5 days [3]. Thus, the probability of emergence following sowing on the i-th DOY is empirically estimated as

$$\hat{p}_i = \frac{\#\{\mathcal{D}_i, \mathcal{D}_{i+1 \bmod 365}\}}{3000} \tag{4}$$

(a) Predictions based on Manhattan distance.



(b) Predictions based on Euclidean distance.

Figure 3: Sowing windows for Brandenburg.

(a) Predictions based on Manhattan distance.

(b) Predictions based on Euclidean distance.

Results

The predicted sowing windows for Brandenburg, Saxony, and Bavaria, obtained using the Manhattan and Euclidean distances, are presented in Figures 3, 4, and 5, respectively. Sowing windows were identified according to the proposed methodology, see Section 3, for which two graph-based approaches identify periods of environmental suitability for maize sowing, although differences are observed in their temporal extent and location. Remarkably, the Manhattan distance yields narrower sowing windows, whereas the Euclidean distance leads to broader periods.

(a) Predictions based on Manhattan distance.

(b) Predictions based on Euclidean distance.

Figure 4: Sowing windows for Saxony.

(a) Predictions based on Manhattan distance.

(b) Predictions based on Euclidean distance.

Figure 5: Sowing windows for Bavaria.

Conclusions

The proposed probabilistic model predicts region-specific sowing windows for Brandenburg, Saxony, and Bavaria, accounting for differences in environmental conditions across regions. The predicted windows during the first half of the year are consistent with the spring-sowing season for maize in Germany [3]. Although both Manhattan and Euclidean distances identify the start of the year, these should be interpreted as early caution, as subsequent viable conditions may limit crop development. Therefore, their practical suitability requires further assessment beyond the sowing-to-emergence period.

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Revista	**Factor de impacto**	**Referencias consultadas**
Computers and Electronics in Agriculture Editorial: Elsevier B.V.	10.3	Elnesr *et al.* (2013)
Sustainable Computing: Informatics and Systems Editorial: Elsevier B.V.	6.2	Gümüşçü *et al.* (2020)
Journal of Agronomy and Crop Science Editorial: Wiley.	3.2	Parker *et al.* (2016)

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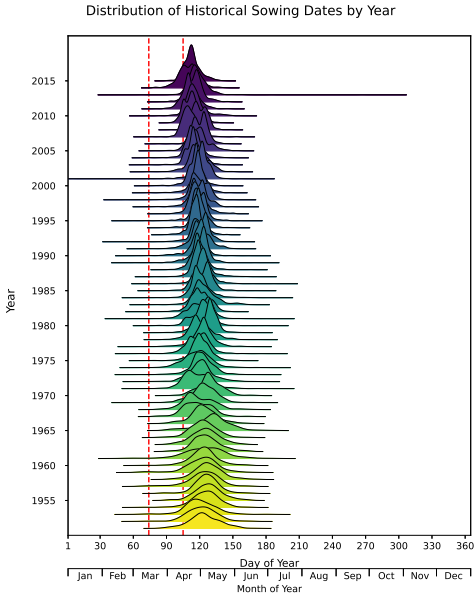
Gracias

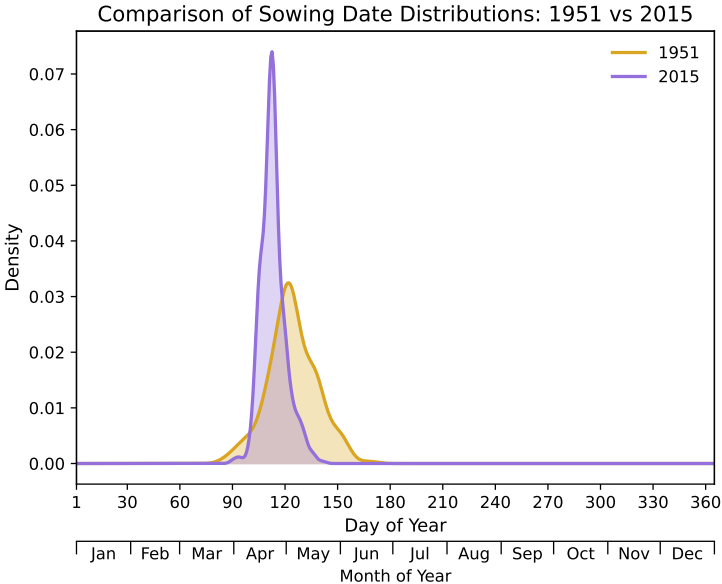
Exploración de la hipótesis



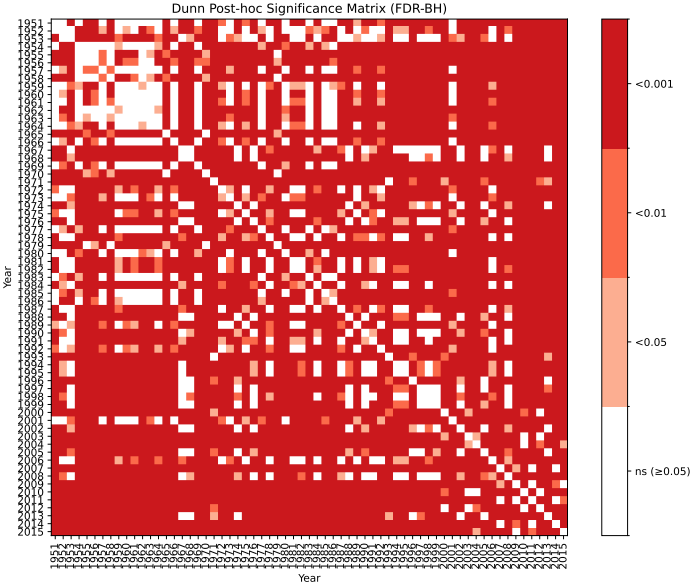
Datos fenológicos

- Calendario agrícola tomado del Joint Research Centre de la Unión Europea.
- Fechas de siembra tomadas de la plataforma Proyecto Fenología Paneuropea 725 (PEP725).
- Periodo 1951–2015.
- Tamaño de la muestra: 58115.

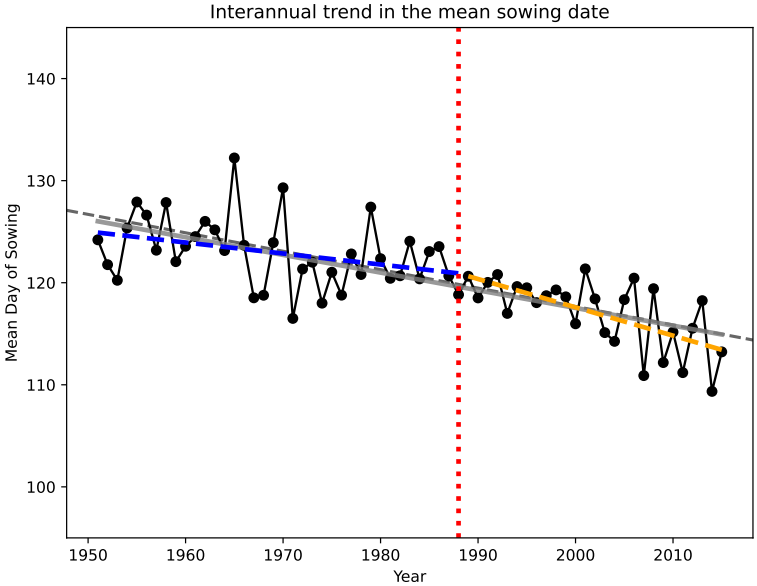
Comparación 1951 vs. 2015



Comparaciones por año

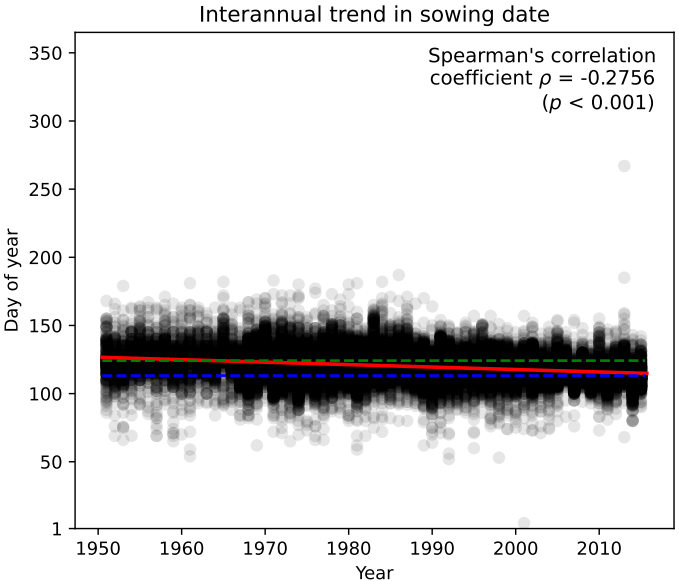


Exploración de la hipótesis

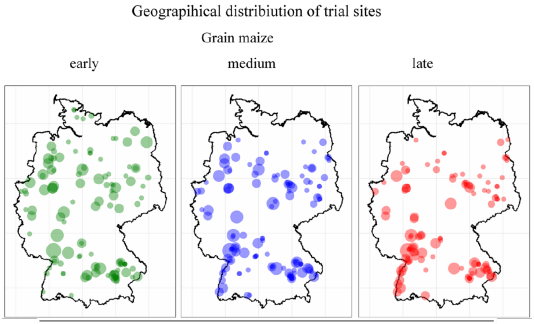


El cambio estructural es consistente con lo previamente reportado por Chmielewski *et al.* (2004); Menzel *et al.* (2020); Reid *et al.* (2015); Wu *et al.* (2016).

Fechas de siembra



Variedades del maíz



Trait	Maturity	First year	Years	Number of Observations	Trials	Varieties	Common varieties
Grain maize							
Grain yield	Early	1987	37	11,218	1,063	186	132
	Medium	1987	37	14,209	1,028	274	179
	Late	1987	37	7,444	854	126	78
Harvest index	Early	1986	35	1,988	269	128	
	Medium	1986	35	3,403	354	175	
	Late	1986	35	2,628	474	77	

Traits	Grain maize				
	Maturity	1987	2023	Diff %	sig
Days from first of January to day of sowing	early	118.1	116.2	−2.0 −1.7	ns
	medium	118.3	115.9	−2.5 −2.1	ns
	late	118.1	116.2	−2.0 −1.7	ns

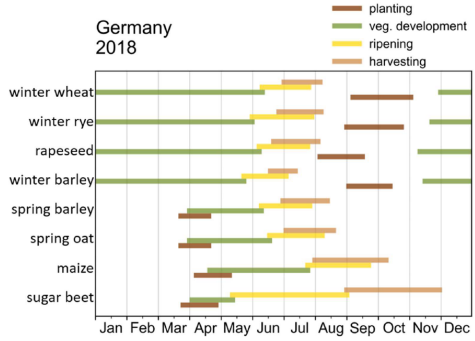
Las distintas variedades de maíz comparten su respuesta a las condiciones de cultivo.

Fuente: Parent *et al.* (2012); Sánchez *et al.* (2014).

Las condiciones climáticas determinan la variedad a emplear.

Fuente: FAO (s.f.).

Fuente: Laidig *et al.* (2025). Breeding progress of grain and forage maize in long-term variety trials compared to on-farm yield development.



Fuente: Asam *et al.* (2022). Mapping Crop Types of Germany by Combining Temporal Statistical Metrics of Sentinel-1 and Sentinel-2 Time Series with LPIS Data.

Quincenas mensuales del año																							
E-I	E-II	F-I	F-II	M-I	M-II	Ab-I	Ab-II	M-I	M-II	Jun-I	Jun-II	Jul-I	Jul-II	Ag-I	Ag-II	S-I	S-II	O-I	O-II	N-I	N-II	D-I	D-II
-	-	-	-	-	1	1	2	2	2	2	2	2	2	2	2	2	3	3	3	3	-	-	-

Fuente: Joint Research Centre of the European Commission. (2026). Agri4cast Data Portal.